Design and Layout of a 2.5 GHz Low Noise Amplifier Using the UMS Process

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***Abstract*— A 2.5 GHz low noise amplifier was developed in Keysight ADS using the UMS process design kit. The design sequence includes transistor geometry and bias selection, wideband stabilization, passive component evaluation, impedance matching, and layout optimization. Removing the dedicated input and output matching networks reduced the layout area and improved the circuit noise figure while retaining useful gain. The compact circuit was back annotated and evaluated with staged RFPro electromagnetic simulations. The final passive EM and active device simulation gives a gain of 11.765 dB at 2.3 GHz and 11.271 dB at 2.7 GHz, with input and output reflection coefficients below approximately -12 dB and -19 dB, respectively, across 2–3 GHz. Its noise figure is approximately 1.14 dB near the 2.5 GHz design frequency.**

***Index Terms*— Low noise amplifier, UMS, GaN HEMT, ADS, source degeneration, RFPro, electromagnetic simulation.**

I. Design Overview

This report describes the design of a 2.5 GHz low noise amplifier using Keysight ADS and the United Monolithic Semiconductors (UMS) process design kit. The initial circuit design follows the UMS LNA tutorial presented by Anurag Bhargava, and the remaining work develops the compact layout and RFPro electromagnetic model. The process consists of selecting the transistor geometry and bias conditions, stabilizing the device, and designing input and output matching networks while accounting for the nonideal behavior of the UMS components.

The amplifier is studied over a frequency range of 2–3 GHz, with a target frequency of 2.5 GHz. The desired criteria were input and output reflection coefficients, S_{11} and S_{22}, below -10 dB, gain above 10 dB, and a noise figure below 1.5 dB. These targets were used to evaluate the gain, noise, and matching throughout the studied band, while stability is examined over a wider frequency sweep. The design was first optimized with input and output matching networks. These networks were then removed to reduce area and passive loss, and the feedback and source degeneration elements were optimized again. The selected topology was placed in layout, back annotated, and evaluated using RFPro. The final results discussed in this report are simulated results from this completed design sequence.

II. Bias and Transistor Geometry

The GH15NHF_NOFP transistor was placed in a DC simulation schematic to obtain its current–voltage characteristics. The GH15 process uses a GaN HEMT, so the bias conditions are defined by the gate–source and drain–source voltages [2]. The initial device consists of four gate fingers with a unit gate width of $75\,\mu\mathrm{m}$. A nested sweep of V_{GS} and V_{DS} was used to identify a suitable operating point.

The drain voltage was selected as $V_{DS} = 5$ V, and the gate voltage was selected as $V_{GS} = -2.4$ V. This places the initial device beyond the knee of the plotted current–voltage curves while maintaining a relatively low drain current. The initial four-finger device draws approximately 25 mA at this operating point. Its displayed DC power consumption is approximately 0.126 W.

The gate width and number of fingers were then swept to see how the geometry affected gain and noise figure. The slider feature in the ADS data display was used to isolate the results for each geometry. After inspection, eight fingers with a unit gate width of $60\,\mu\mathrm{m}$ were selected as a suitable compromise between gain and noise figure.

The selected device has a total gate width of $8(60\,\mu\mathrm{m}) = 0.480$ mm. The displayed drain current is 40.298 mA, corresponding to approximately 84.0 mA/mm. This falls within the recommended 50–150 mA/mm range shown in the model documentation. The geometry study gives 17.043 dB for S_{21} at 2.5 GHz before the stabilization and matching networks are introduced.

The geometry selection provided a starting point for the rest of the design. Some of the available gain would be lost when feedback and the physical passive components were added, so it was useful to retain gain margin at this stage.

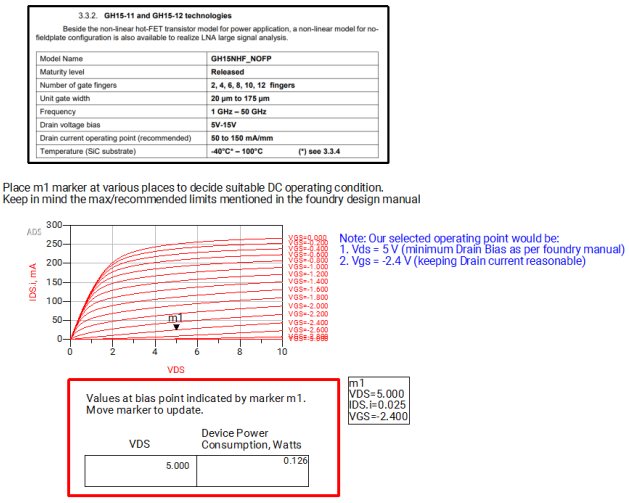


Fig. 1. Foundry model limits and the selected DC operating point for the initial transistor geometry.

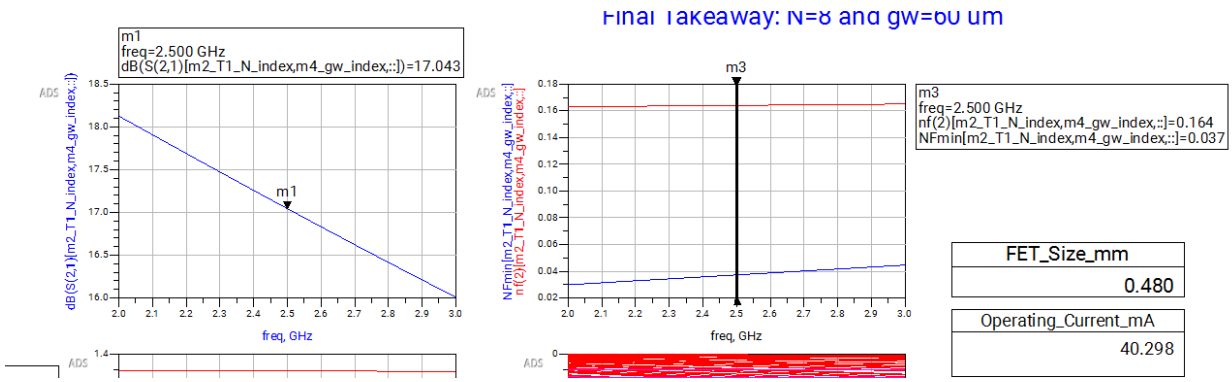


Fig. 2. Isolated gain, noise, and operating current results for eight fingers and a unit gate width of 60 μm.

III. Stabilization

After the geometry and bias conditions were selected, stability analysis was conducted on the transistor. A drain-to-gate RLC feedback network and two inductive source-to-ground branches were added to the schematic. The feedback network was optimized to improve stability while preserving useful gain and a low noise figure. The two source branches provide source degeneration through the inductors and their ground connections. The transistor artwork was also inspected before placing the complete amplifier layout.

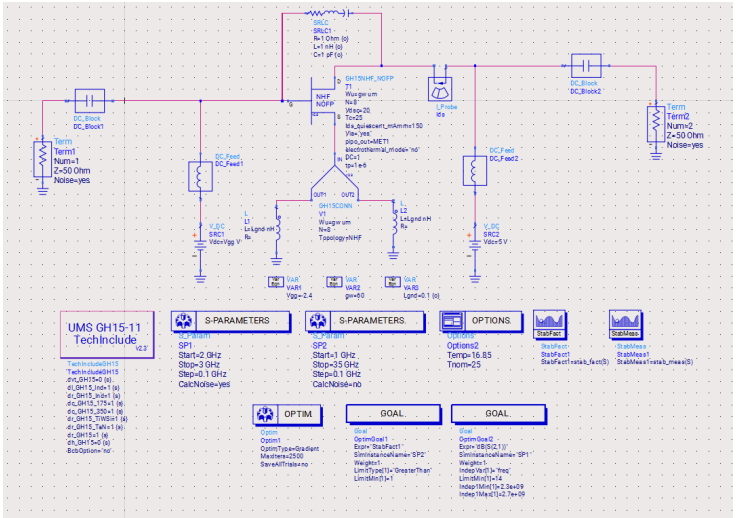


Fig. 3. Initial stabilization schematic with ideal DC feeds and blocks, an RLC feedback network, and source degeneration inductance.

The stability factor K and stability measure B_1 were evaluated together [1]. They are defined as

$$\Delta = S_{11}S_{22} - S_{12}S_{21}, \tag{1}$$

$$K = \frac{1 - |S_{11}|^2 - |S_{22}|^2 + |\Delta|^2}{2|S_{12}S_{21}|}, \tag{2}$$

$$B_1 = 1 + |S_{11}|^2 - |S_{22}|^2 - |\Delta|^2. \tag{3}$$

For this analysis, unconditional small signal stability requires $K > 1$ and $B_1 > 0$. The post-optimization results satisfy these conditions over the plotted sweep to 35 GHz. The minimum displayed K is approximately 1.005, indicating that the stability margin is small at the upper end of the sweep. These results describe the simulated circuit and frequency range.

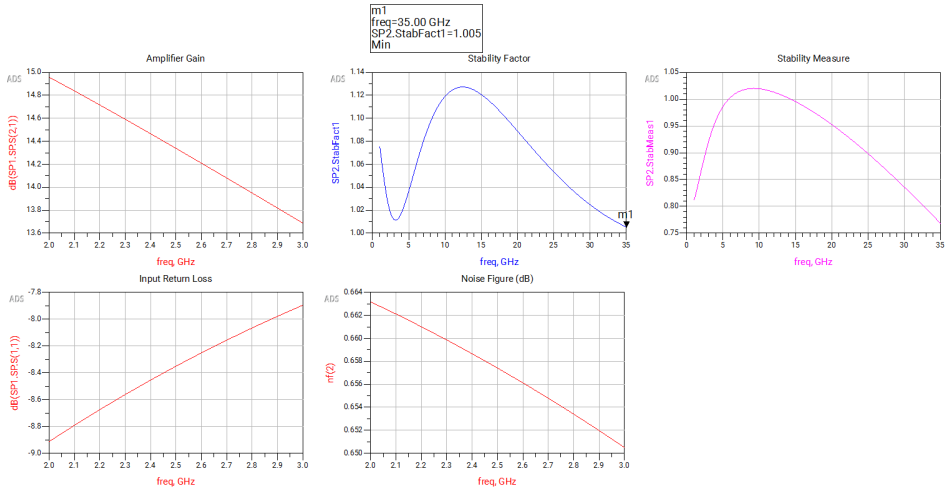


Fig. 4. Gain, input reflection coefficient, noise figure, and stability after optimization of the ideal stabilization network.

Available gain circles and noise figure circles were then plotted using measurement components from the ADS library. These circles provide a way to inspect the relationship between source impedance, gain, and noise figure before implementing a matching network.

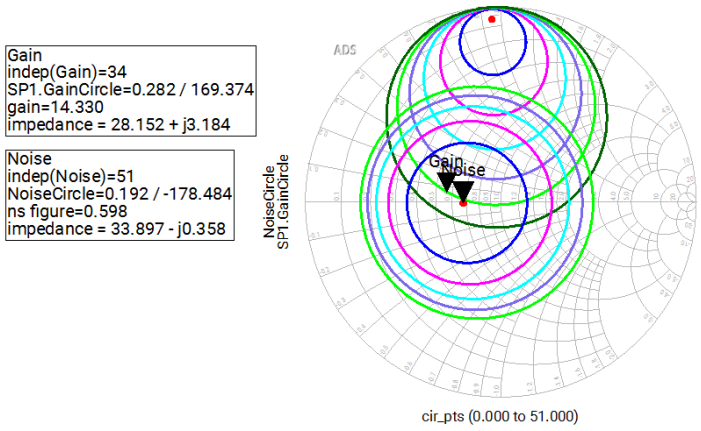


Fig. 5. Available gain and noise figure circles for the initially stabilized device.

The displayed gain-circle marker indicates approximately 14.3 dB, and the noise-circle marker indicates approximately 0.6 dB. The markers correspond to slightly different impedances, so they describe the nearby gain and noise tradeoff rather than one exact operating point. The initial optimum noise source impedance is approximately $(33.897 - j0.358)\,\Omega$.

IV. UMS Passive Components

The initial analysis uses ideal DC feed and DC block components. These were replaced with spiral inductors, capacitors, and resistors from the UMS design kit to account for their frequency dependent behavior. Explicit via models were also included at the ground connections.

At 2.5 GHz, an inductive reactance of approximately $85\,\Omega$ and a capacitive reactance magnitude of approximately $5\,\Omega$ were used as starting values for the RF chokes and DC blocking capacitors. The corresponding ideal values are

$$L = \frac{X_L}{2\pi f_0} = \frac{85}{2\pi(2.5\times 10^9)} \approx 5.41\,\text{nH}, \tag{4}$$

$$C = \frac{1}{2\pi f_0 |X_C|} = \frac{1}{2\pi(2.5\times 10^9)(5)} \approx 12.73\,\text{pF}. \tag{5}$$

Nominal values of approximately 5.5 nH and 12.8 pF were used in the UMS implementation. These values provide an initial estimate; the actual component response is determined by the PDK models.

Replacing the ideal components with UMS components reduced the stability margin, with K falling below unity over part of the sweep. This demonstrates that the stabilization obtained with ideal components does not necessarily carry over when their nonideal equivalents are introduced.

The source degeneration inductors and feedback network were then optimized again. The resulting schematic includes the UMS via models and the updated stabilization components.

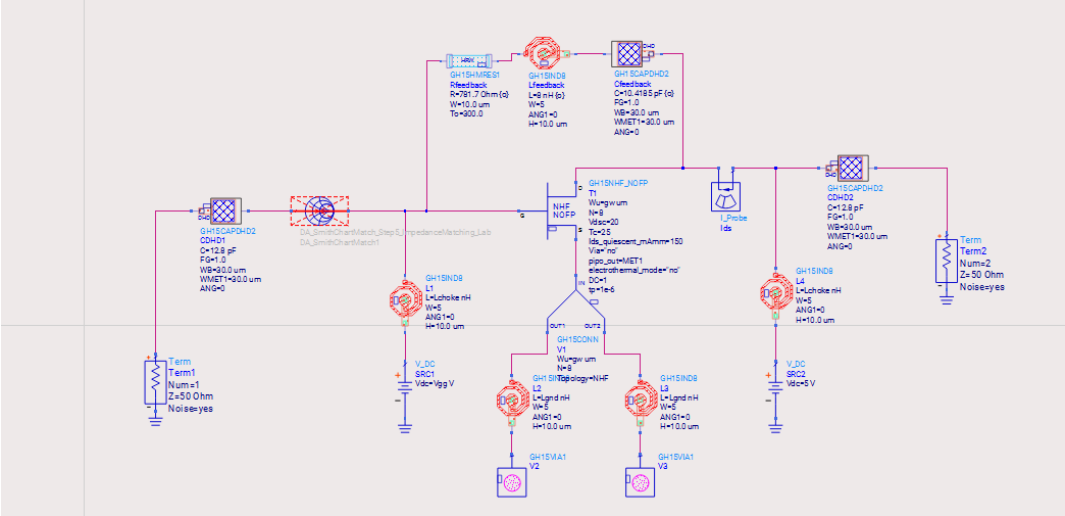


Fig. 6. Updated UMS schematic after optimization of the stabilization network and ground connections.

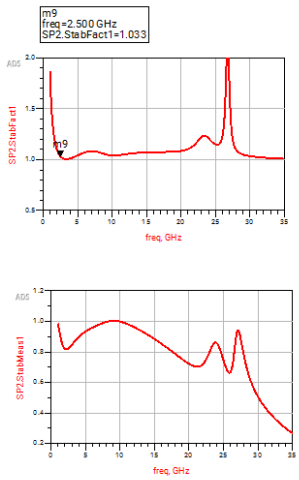


Fig. 7. Stability response after reoptimization, with $K = 1.033$ at 2.5 GHz and a positive stability measure over the plotted sweep.

After reoptimization, the plots indicate unconditional stability over the displayed frequency range. The input and output matching impedances now had to be determined again because the stabilization network and passive models changed the response of the amplifier.

A. Passive Component Response

The spiral inductor and MIM capacitor were studied separately to examine the difference between their nominal values and their effective values at the operating frequency. Their impedance was obtained from the simulated reflection coefficient, and the reactive part was used to calculate effective inductance or capacitance.

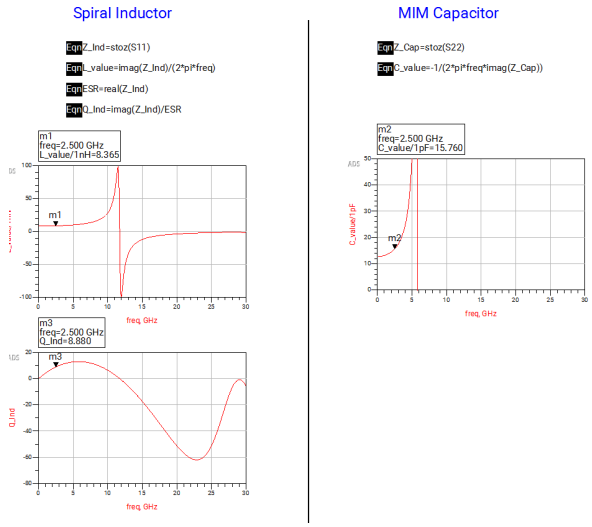


Fig. 8. Frequency response of the effective inductance, inductor quality factor, and effective capacitance.

At 2.5 GHz, the nominal 8 nH inductor produces an effective inductance of approximately 8.365 nH and a quality factor of 8.880. The nominal 12.8 pF capacitor produces an effective capacitance of approximately 15.760 pF. While the response is usable at the operating frequency, the difference from the nominal values explains why replacing ideal components changes the matching and noise performance.

This was a useful check because the nominal component value alone does not describe its response at 2.5 GHz. The matching networks therefore had to be evaluated with the UMS models before deciding which component values to keep.

V. Statistical Analysis

Monte Carlo simulation was performed to study the effect of component tolerances before the input and output matching networks were implemented. The resulting traces show variation in the input and output reflection coefficients, gain, and noise figure.

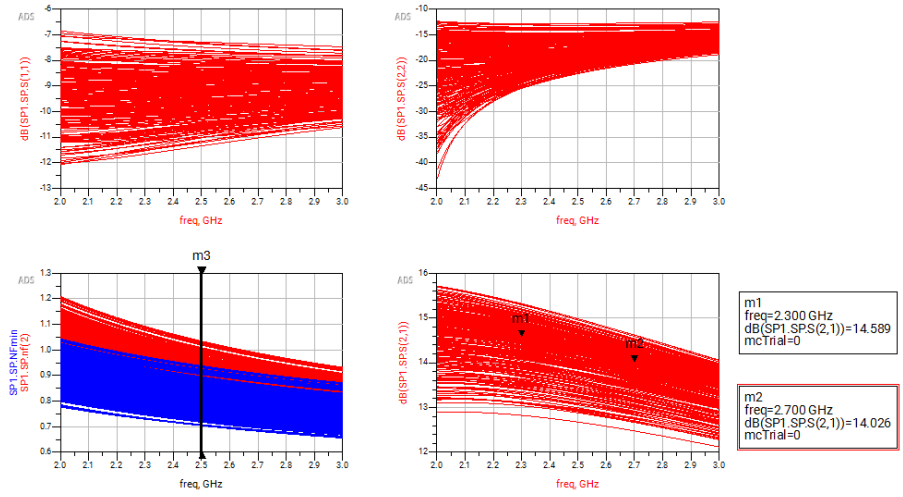


Fig. 9. Monte Carlo results before implementation of the input and output matching networks.

The matching characteristics show a substantial amount of variation between trials. This indicates that the nominal matching solution should include enough margin to account for changes in component values. The plotted results describe the variation at this stage of the design. They do not establish a yield estimate for the later compact layout or EM model.

VI. Impedance Matching

A. *Input Matching*

After the UMS component behavior was examined, the input matching network was designed using the ADS Smith Chart Utility. The initial input network was selected to present the optimum noise source impedance to the stabilized amplifier. With $Z_0 = 50\,\Omega$, the displayed noise impedance is

$$Z_{S,\mathrm{opt}} = 50(0.620 - j0.084) = (31 - j4.2)\,\Omega. \tag{6}$$

The Smith Chart Utility uses the conjugate impedance $(31 + j4.2)\,\Omega$ as the load for synthesizing the lossless network. When connected ahead of the amplifier and terminated by the 50 Ω source, the network presents the desired source impedance to the amplifier.

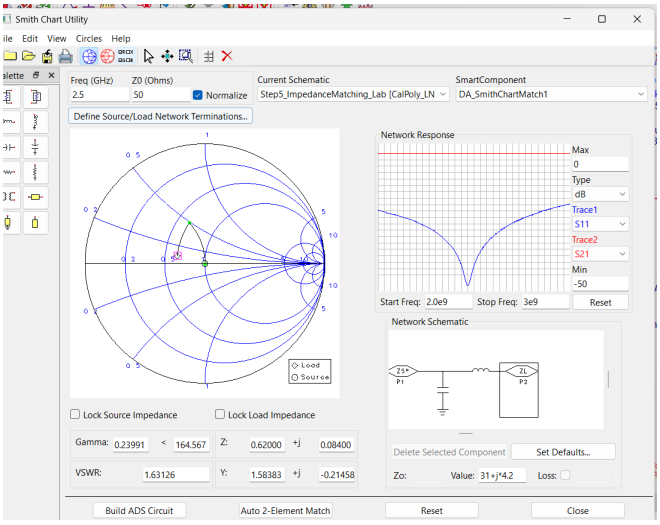


Fig. 10. ADS Smith Chart Utility used to synthesize the initial input matching network.

The ideal network produces a noise figure close to the displayed minimum, approximately 0.798 dB at 2.5 GHz. However, the input reflection coefficient remains approximately -5.8 dB near the target frequency. Minimum noise is obtained when the source reflection coefficient approaches Γ_{opt}, whereas a power match requires the conjugate of the amplifier input reflection coefficient [1]. These conditions generally differ, which explains the poor input match even though the noise figure is minimized.

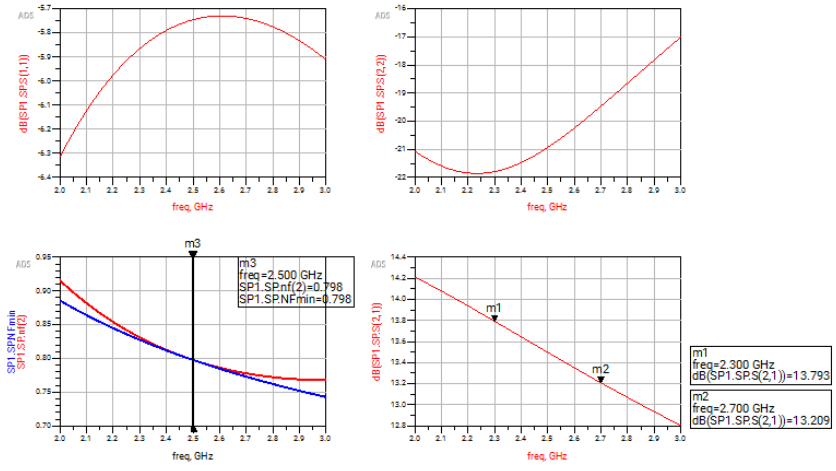


Fig. 11. Amplifier response with the ideal input network selected for minimum noise.

The ideal network was replaced with a UMS series inductor, a shunt capacitor, and a ground via. This provides a physical implementation of the input matching topology while including the component model nonidealities.

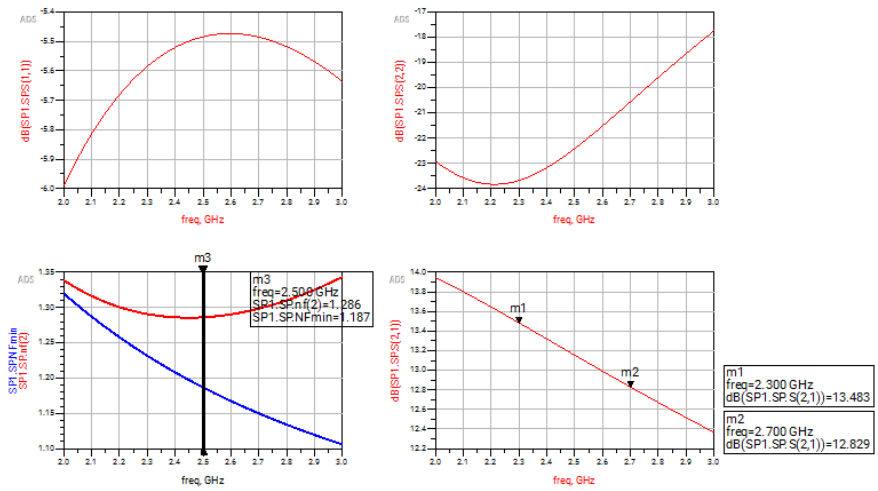


Fig. 12. Amplifier response after replacement of the ideal input matching elements with UMS components.

At this stage, the displayed noise figure rises to 1.286 dB at 2.5 GHz, compared with a minimum of 1.187 dB for the updated circuit. The input reflection coefficient remains above the desired -10 dB level. Further optimization is therefore required to obtain acceptable matching while preserving a low noise figure.

B. Output Matching

The output matching network was designed using the same process. The displayed output reflection coefficient corresponds to an approximate output impedance of

$$Z_{\text{out}} = 50(1.163 - j0.016) = (58.15 - j0.8)\,\Omega. \tag{7}$$

The desired load is its conjugate, approximately $(58.15 + j0.8)\,\Omega$, for the source termination used in this simulation.

The ideal LC network produces a deep minimum in the output reflection coefficient near 2.5 GHz. The network was then replaced with a UMS series inductor, a shunt capacitor, and a ground via.

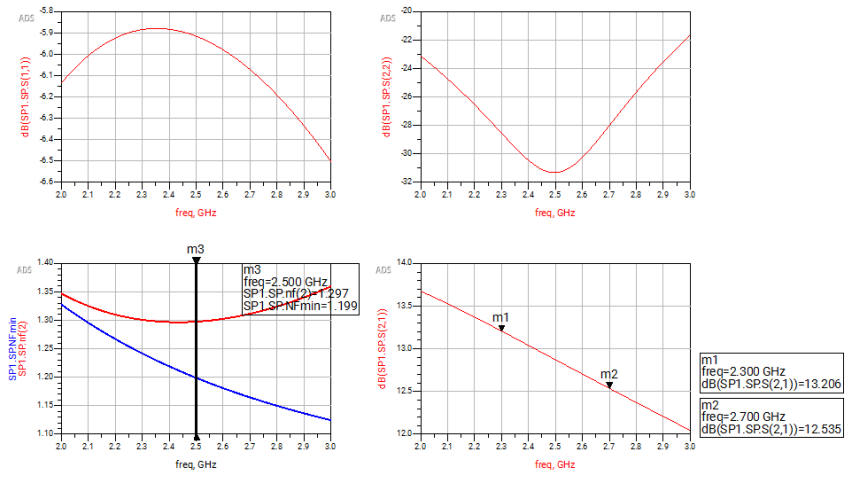


Fig. 13. Amplifier response after implementation of the UMS output matching network.

The nonideal implementation reduces the depth of the output reflection minimum, but the output matching remains suitable over the studied band. The input and output matching networks and stabilization components were subsequently optimized together to improve the overall gain, noise figure, and matching response.

VII. Matching Network Tradeoff

The initial optimized circuit includes both matching networks and the UMS stabilization components. Its response is shown in Fig. 14. The gain reaches approximately 13.8 dB, with markers of 13.772 dB at 2.3 GHz and 13.516 dB at 2.7 GHz. Both reflection coefficients remain below -10 dB across 2–3 GHz. At 2.5 GHz, the noise figure is 1.092 dB, compared with the circuit minimum of 0.953 dB.

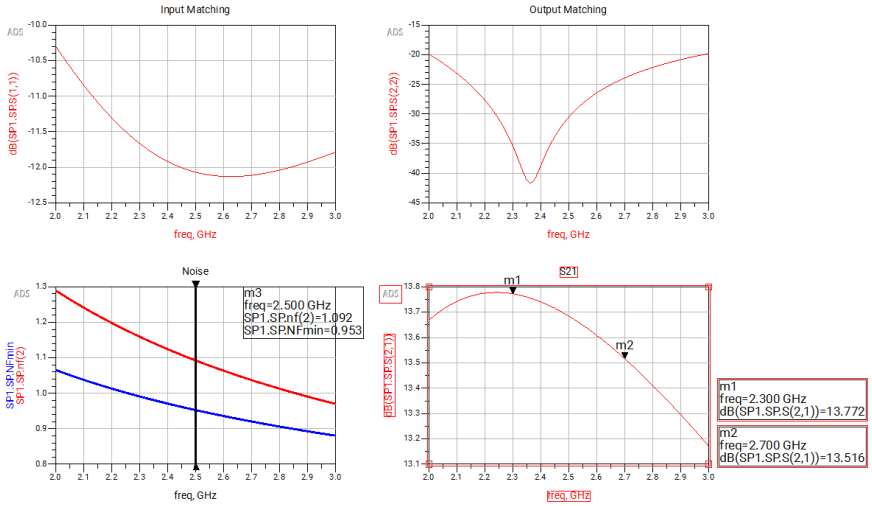


Fig. 14. Response with both matching networks. Noise markers are at 2.5 GHz.

While this circuit provides good gain and matching, the input network introduces passive loss ahead of the transistor. It was removed to determine whether the feedback and source degeneration elements could provide a suitable response on their own. The remaining components were then optimized again.

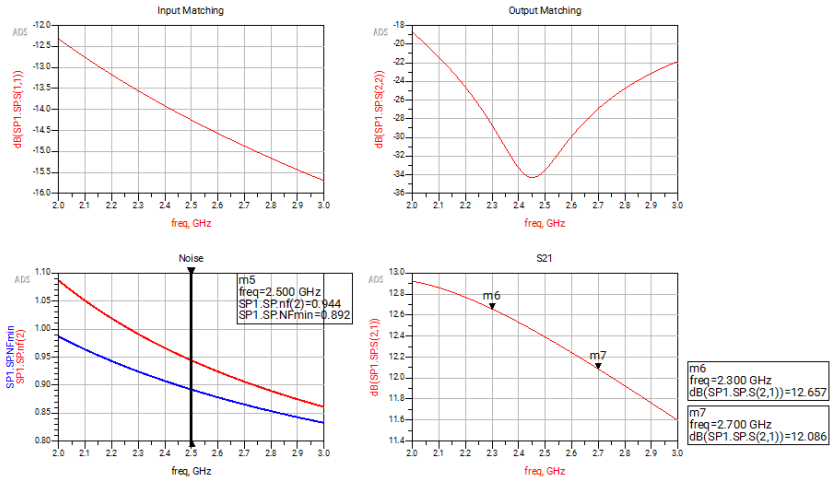


Fig. 15. Response with only the dedicated output matching network.

Removing the input network lowered the 2.5 GHz noise figure to 0.944 dB, compared with a minimum of 0.892 dB. However, gain decreased to 12.657 dB at 2.3 GHz and 12.086 dB at 2.7 GHz. This is about 1.1–1.4 dB below the original matched circuit at the two marker frequencies. The lower noise figure was useful, but there was still an opportunity to reduce the component count and recover some gain.

The output matching network was then removed as well, and the feedback and source degeneration components were optimized together. The selected circuit retains the RF chokes, DC blocks, feedback path, and source degeneration elements, but has no separate input or output matching network.

The compact circuit provides 13.292 dB of gain at 2.3 GHz and 12.855 dB at 2.7 GHz. Its noise figure is 0.986 dB at 2.5 GHz, with a minimum of 0.909 dB. The input reflection coefficient is around -13 to -14 dB, while the output reflection coefficient is below approximately -24 dB across 2–3 GHz. Relative to the original matched design, the gain is lower by 0.48 dB at 2.3 GHz and

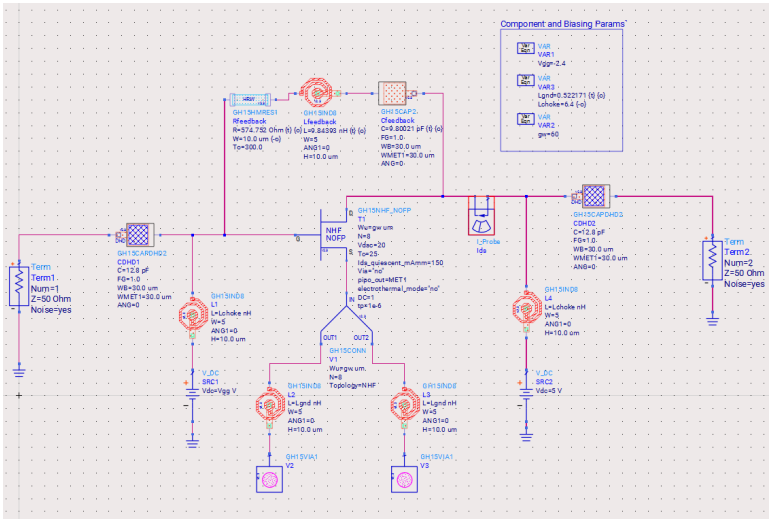


Fig. 16. Selected compact topology after removal of both dedicated matching networks.

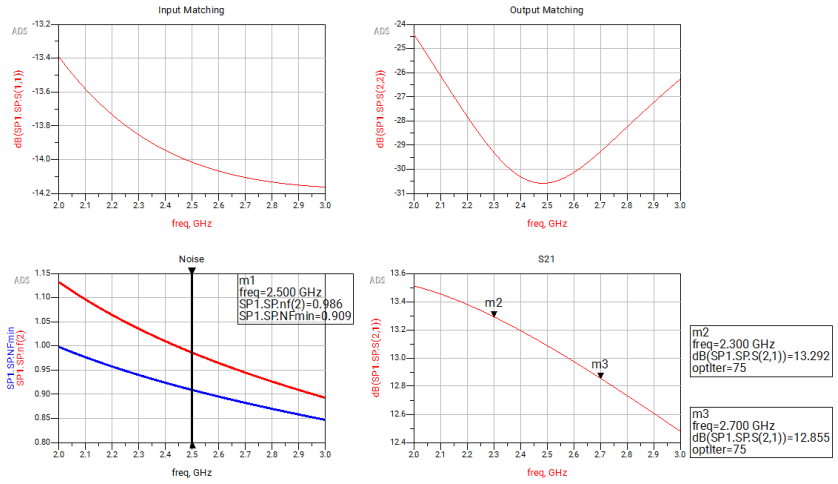


Fig. 17. Response of the compact circuit before layout interconnects are included.

0.66 dB at 2.7 GHz, while the 2.5 GHz noise figure is lower by 0.106 dB. This provided a useful compromise between noise, gain, matching, and the number of components that must fit in layout.

The lower noise figure and smaller component count made the gain reduction acceptable for this design. The feedback and source degeneration elements already provided useful matching, so the circuit could be simplified before moving into layout.

TABLE I
Circuit results during topology and interconnect optimization. All values are in dB.

Design stage	S_{21} 2.3 GHz	S_{21} 2.7 GHz	NF 2.5 GHz	$\mathrm{NF}_{\mathrm{min}}$ 2.5 GHz
Both matching networks	13.772	13.516	1.092	0.953
Output matching network only	12.657	12.086	0.944	0.892
Neither matching network	13.292	12.855	0.986	0.909
Initial back annotation	13.172	12.733	0.983	0.903
Optimized back annotation	13.538	13.143	0.974	0.887

VIII. Layout and Back Annotation

The layout was developed around a $1\,\mathrm{mm}^2$ area target. The PDK components were populated in layout, then their positions and interconnects were adjusted to form a compact arrangement. Removing the dedicated matching networks made this step more manageable because fewer inductors, capacitors, and ground connections had to be placed. The initial arrangement was then refined during interconnect optimization.

The layout was back annotated into the schematic, adding transmission line representations of the connections between components. This gives an intermediate model between the original circuit schematic and the later EM simulation. The added connections are shown in Fig. 18.

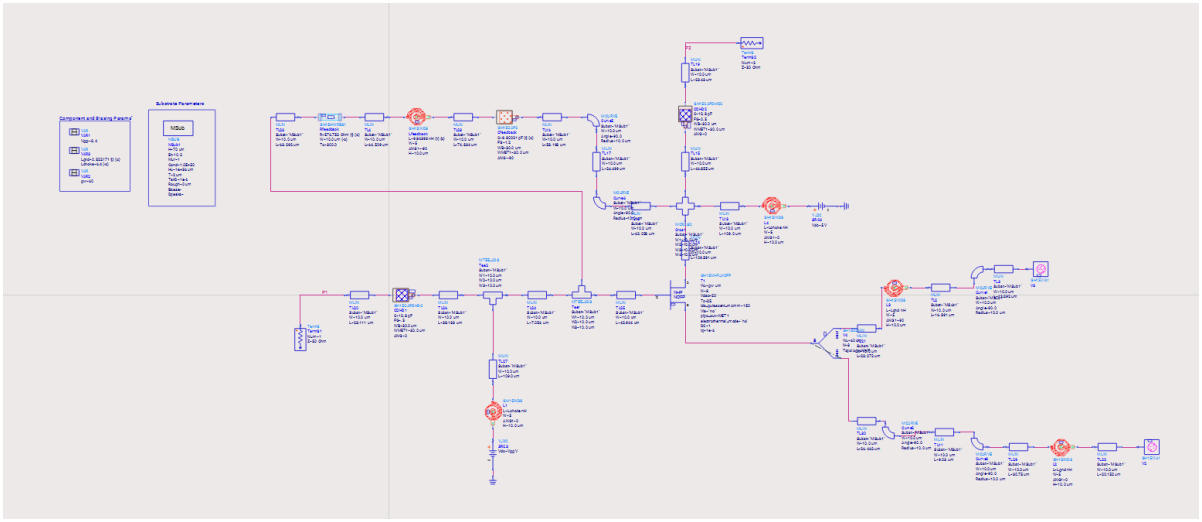


Fig. 18. Back annotated schematic with transmission line models for the layout interconnects.

The initial back annotation produced results close to the compact circuit simulation. Gain changed to 13.172 dB at 2.3 GHz and 12.733 dB at 2.7 GHz, about 0.12 dB lower at each marker. The noise figure changed only slightly, to 0.983 dB at 2.5 GHz. The first routing arrangement was therefore a reasonable starting point, although its dimensions could still be adjusted to improve the response.

The connections near the gate, drain, and source form part of the RF circuit, even when they look like short pieces of routing. It was therefore useful to evaluate their widths while the layout could still be adjusted.

Four interconnect widths were included in the next optimization: the feedback path, gate input connection, drain output connection, and source degeneration connection. The PDK component values were also adjusted. Figure 19 shows the optimized response.

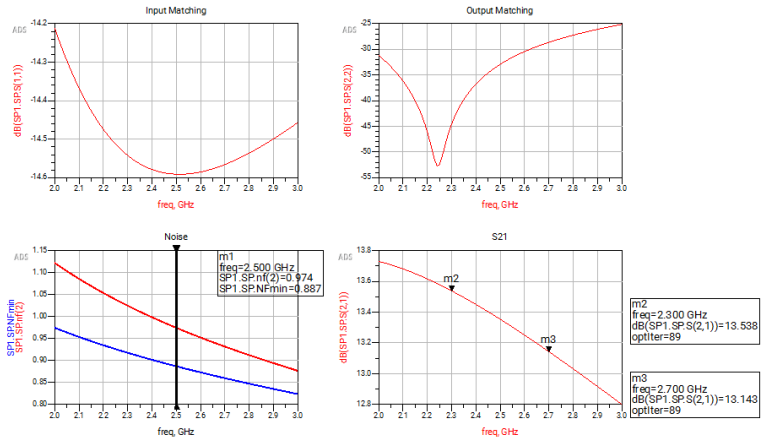


Fig. 19. Back annotated response after optimization of the interconnect widths and PDK components.

After optimization, gain increased to 13.538 dB at 2.3 GHz and 13.143 dB at 2.7 GHz. The noise figure decreased to 0.974 dB at 2.5 GHz, with a minimum of 0.887 dB. The input reflection coefficient stayed near -14 dB, while the output response developed a

deeper matching minimum. The updated values and routing dimensions were transferred back to layout. The updated layout passed the PDK design rule check.

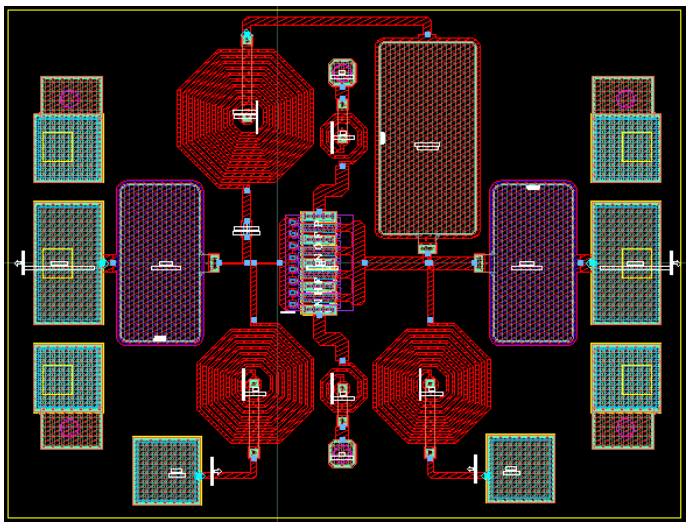


Fig. 20. Updated layout with pads and the optimized routing arrangement, used for RFPro analysis.

Back annotation shows how the physical connections affect the circuit response. However, transmission line models do not represent all of the coupling and distributed behavior of the full passive layout. These effects were examined using RFPro.

IX. RFPro Electromagnetic Simulation

A. *Interconnect Extraction*

The first RFPro analysis included the interconnects, pads, and vias in the EM model while retaining circuit models for the other components. This provided a comparison with the back annotated response. In Fig. 21, the blue traces represent the extracted response and the red traces represent the circuit comparison.

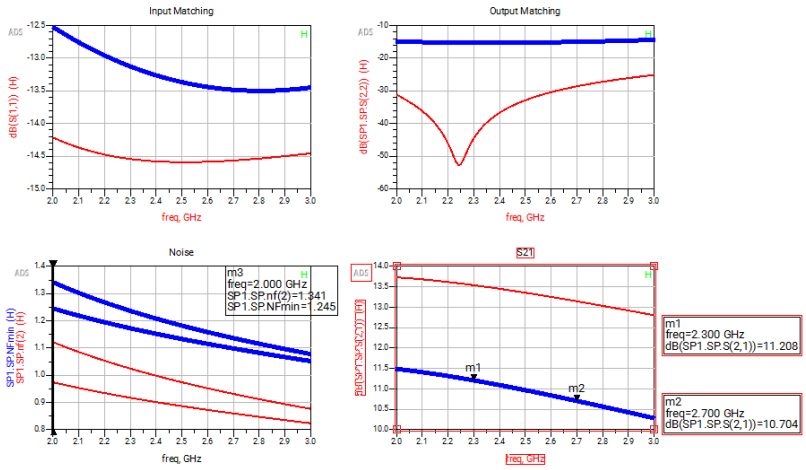


Fig. 21. Interconnect, pad, and via EM response compared with the circuit model. The noise markers are at 2.0 GHz.

The extracted response showed a noticeable reduction in gain and a shallower output match. The EM gain markers are 11.208 dB at 2.3 GHz and 10.704 dB at 2.7 GHz, about 2.3–2.4 dB below the optimized back annotated circuit. The output reflection coefficient is close to -15 dB over much of the plotted band. The small change seen during the first back annotation therefore did not capture the entire layout effect.

This result shows that simulation based on the back annotated schematic alone does not sufficiently describe the parasitics present within the layout. DRC checks whether the geometry satisfies the process rules, while the RFPro analysis accounts for its effect on the modeled gain and matching.

At the displayed 2.0 GHz marker, the noise figure is 1.341 dB and the minimum noise figure is 1.245 dB. These values should not be directly compared with the 2.5 GHz circuit values in Table I, since they occur at different frequencies.

B. *Full Passive Layout Extraction*

The next analysis extended the EM partition to include the passive layout structures, including the inductors, resistors, capacitors, and interconnects. The transistor remained represented by its active device circuit model. Figure 22 shows the passive layout used for the extraction, and Fig. 23 shows the final response.

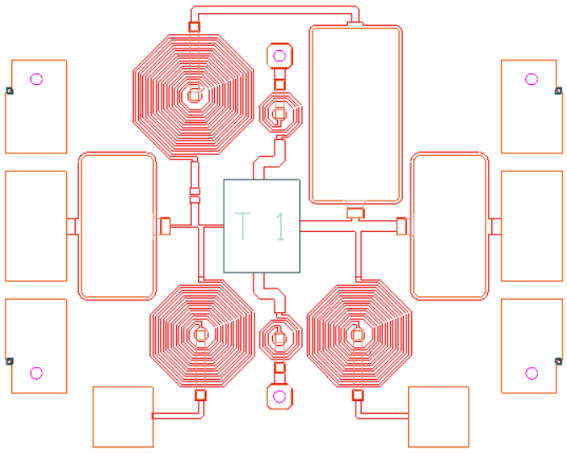


Fig. 22. Passive layout partition for the final RFPro extraction. The active transistor is retained as a circuit model.

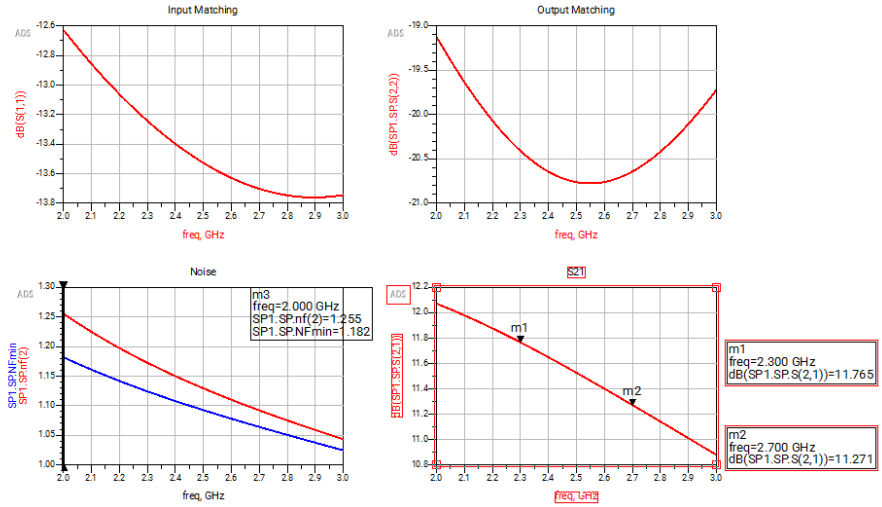


Fig. 23. Final full passive EM and active device circuit response. The noise markers are at 2.0 GHz.

The final extraction improved the response relative to the interconnect-only case. Gain increased to 11.765 dB at 2.3 GHz and 11.271 dB at 2.7 GHz, an improvement of about 0.56 dB at each marker. The output reflection coefficient improved to approximately -19 to -21 dB, while the input reflection coefficient remained around -13 dB across 2–3 GHz. Both ports therefore retain acceptable matching over the studied band.

At 2.0 GHz, the final noise figure is 1.255 dB and the minimum noise figure is 1.182 dB. Their difference is only 0.073 dB, indicating that the source termination is close to the noise optimum of the extracted circuit at this frequency. Near the 2.5 GHz design frequency, the plotted noise figure is approximately 1.14 dB and the minimum is approximately 1.08 dB. The 2.5 GHz values are estimates read from the curves rather than exact marker values.

TABLE II
Selected values from the RFPro simulation stages. All values are in dB.

EM stage	S_{21} 2.3 GHz	S_{21} 2.7 GHz	NF 2.0 GHz	$\mathrm{NF}_{\mathrm{min}}$ 2.0 GHz
Interconnects, pads, and vias	11.208	10.704	1.341	1.245
Full passive layout	11.765	11.271	1.255	1.182

The final gain is still about 1.8–1.9 dB below the optimized back annotated result at the two marker frequencies. Extending the EM partition changes the modeled response and improves gain relative to the interconnect-only extraction, but does not recover the original circuit gain. The full passive extraction was therefore used for the final assessment of the amplifier.

X. Discussion and Conclusion

The completed design shows why matching, stabilization, and layout must be considered together. The original matched circuit produced the highest gain of the compared circuit topologies, but its additional passive components increased noise figure and layout area. Removing both matching networks allowed the feedback and source degeneration elements to provide the required matching with fewer components. This became the selected topology because it retained most of the circuit gain while reducing noise figure relative to the original matched design.

Back annotation initially changed the response only slightly, and interconnect optimization improved gain and noise figure further. RFPro then produced a larger change, particularly in gain and output matching. With the full passive layout extracted, the amplifier retained input and output reflection coefficients below approximately -12 dB and -19 dB across 2–3 GHz, respectively. Its gain is roughly 11–12 dB over this range, and its noise figure is close to the minimum of the extracted circuit. The final simulated response meets all three design criteria: input and output matching below -10 dB, gain above 10 dB, and noise figure below 1.5 dB.

The earlier wideband K and B_1 plots document the stabilization stages. The supplied final EM plots show gain, matching, and noise, but do not include a final wideband stability sweep. They also do not provide a final layout Monte Carlo yield result or a large signal linearity assessment. The report therefore uses those earlier studies to describe the design process rather than as verification of the completed EM model.

Overall, the final circuit provides a low noise response, useful matching, and moderate gain in a compact topology. This combination makes it a reasonable candidate for a receiver preamplifier, where noise performance is important and additional gain can be provided by later stages. The completed layout and EM results support this conclusion more directly than the initial schematic response alone.

References

[1] Keysight Technologies, *Practical RF Amplifier Design Using the Available Gain Procedure and the Advanced Design System EM*, application note 5990-3356, pp. 5–8. [Online]. Available: `https://www.keysight.com/content/dam/keysight/en/doc/gate/white-papers/5990-3356.pdf`.

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